

Shiftless Collatz Model:

Binary Interference in the $3n+LSB$ Trajectory

Hiroshi Harada — March 21, 2026

1. Introduction

The classical Collatz conjecture applies the mapping “ $3n+1$ ” to odd integers and “ $n/2$ ” to even integers. However, the division by 2 is a right-shift operation that physically discards the least significant bit (LSB), thereby erasing information from the system. This study focuses on this information loss and proposes a new mapping, the Shiftless Collatz model, which completely eliminates the right-shift operation.

The Shiftless Collatz model uses the following recurrence:

- $n_{k+1} = 3 \cdot n_k + L_k$
- $L_k = 2^v$

Here, v is the position of the least significant 1-bit of n_k .

By treating the LSB as its full binary weight (2^v) rather than a Boolean 0/1, the Shiftless Collatz model preserves all bit-level information and allows the trajectory to evolve without losing any binary structure.

This study analyzes the trajectory of the Shiftless Collatz model and clarifies the following properties:

- Equivalence to the standard Collatz model
- Emergence of fractal structures
- Linear decomposition and interference geometry

2. Equivalence to the Standard Collatz Model

In the standard Collatz model, the odd-step transformation can be decomposed as:

- $3n + 1 = 2^k \cdot m$

and the next state is the odd part m .

In the Shiftless Collatz model, we apply:

- $n_{k+1} = 3n_k + LSB \cdot n_k$

For the initial value $n_0 = 27$, the following facts were confirmed (**Figure 1**):

- 1) The odd part of n_k (after removing all factors of 2) matches exactly the odd part in the standard Collatz sequence.
- 2) The number of steps required for n_k to reach a power of two (the “jackpot”) matches the number of odd steps required for the standard Collatz sequence to reach 1.

These results show that the Shiftless Collatz model is mathematically equivalent to the standard Collatz model.

Verification for Seed: 27

Step	Standard (Odd)	Shiftless	Odd(Shiftless)	Match
1	41	82	41	True
2	31	248	31	True
3	47	752	47	True
4	71	2272	71	True
5	107	6848	107	True
6	161	20608	161	True
7	121	61952	121	True
8	91	186368	91	True
9	137	561152	137	True
10	103	1687552	103	True
11	155	5079040	155	True
12	233	15269888	233	True
13	175	45875200	175	True
14	263	137887744	263	True
15	395	414187520	395	True
16	593	1243611136	593	True
17	445	3732930560	445	True
18	167	11207180288	167	True
19	251	33688649728	251	True
20	377	101200166912	377	True
21	283	303868936192	283	True
22	425	912680550400	425	True
23	319	2740189134848	319	True
24	479	8229157339136	479	True
25	719	24704651886592	719	True
26	1079	74148315398144	1079	True
27	1619	222513665671168	1619	True
28	2429	667678435966976	2429	True
29	911	2003310185807872	911	True
30	1367	6012129580679168	1367	True
31	2051	18040786788548608	2051	True
32	3077	54131156458668032	3077	True
33	577	162411061562048512	577	True
34	433	487514659662856192	433	True
35	325	1463669878895411200	325	True
36	61	4395513236313604096	61	True
37	23	13258597302978740224	23	True
38	35	40352252661239644160	35	True
39	53	122209679488325779456	53	True
40	5	368934881474191032320	5	True
41	1	1180591620717411303424	1	True

Verification Successful. Total steps to Jackpot: 41
Final Jackpot Value: 2^{70}

Figure 1. Comparison of the odd parts of the standard Collatz sequence and the Shiftless Collatz sequence ($n = 27$).

3. Visualization of the Shiftless Collatz Trajectory

In the Shiftless Collatz model, the injected LSB triggers cascading carries in the lower bits, producing the following characteristic behaviors:

- Lower bits rapidly vanish
- Upper bits exhibit Sierpiński-gasket-like structures
- The overall trajectory forms a complex fractal pattern

Figure 2 visualizes the coexistence of the carry avalanche and the gasket structure.

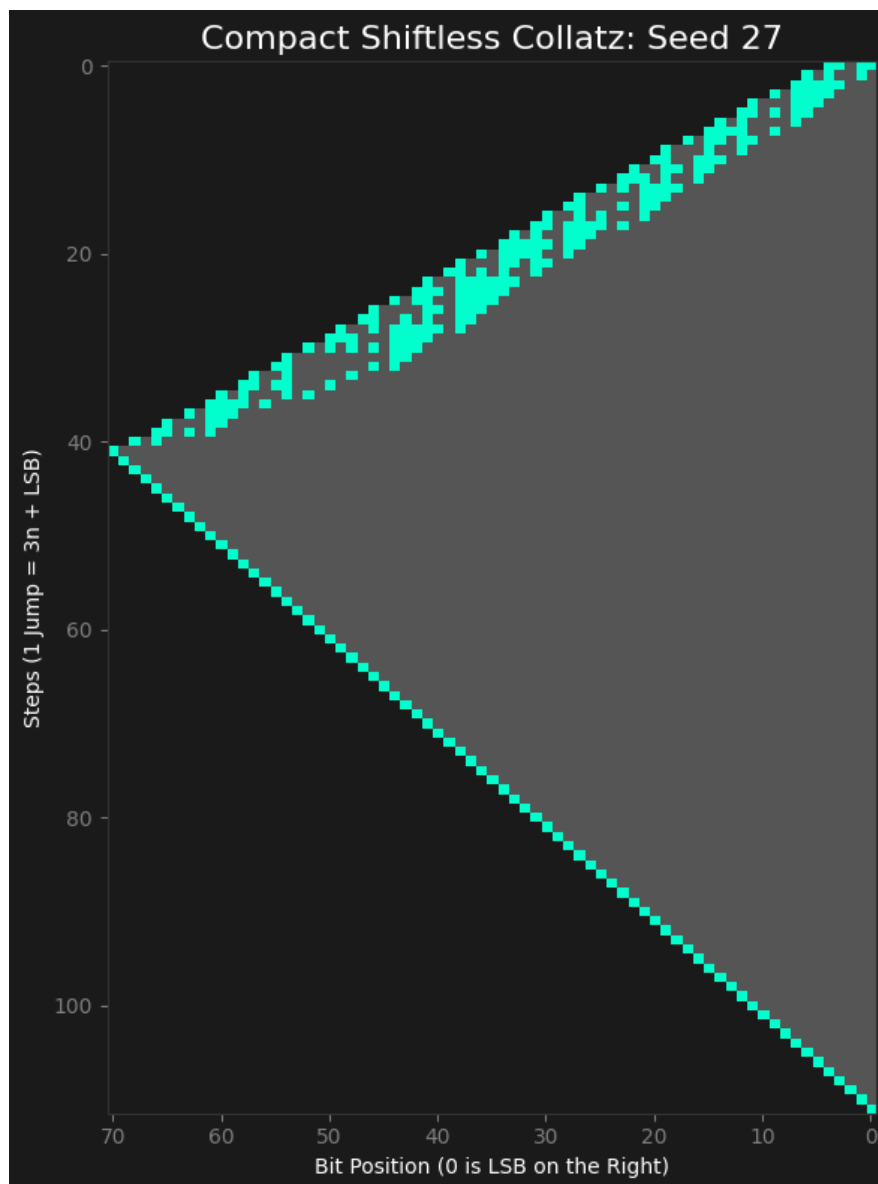


Figure 2. Compact visualization of the Shiftless Collatz trajectory ($n = 27$).

4. Two-Phase Structure of the Shiftless Collatz Step

Each step of the Shiftless Collatz model can be decomposed into two phases:

- 1) Reach phase:

$$n \rightarrow 3n$$

This self-overlapping operation generates Pascal-triangle-like and Sierpiński-gasket-like structures.

- 2) Fill phase:

$$3n \rightarrow 3n + LSB$$

The injected LSB introduces local distortions and triggers carry avalanches.

Figure 3 visualizes this internal Reach-and-Fill structure.

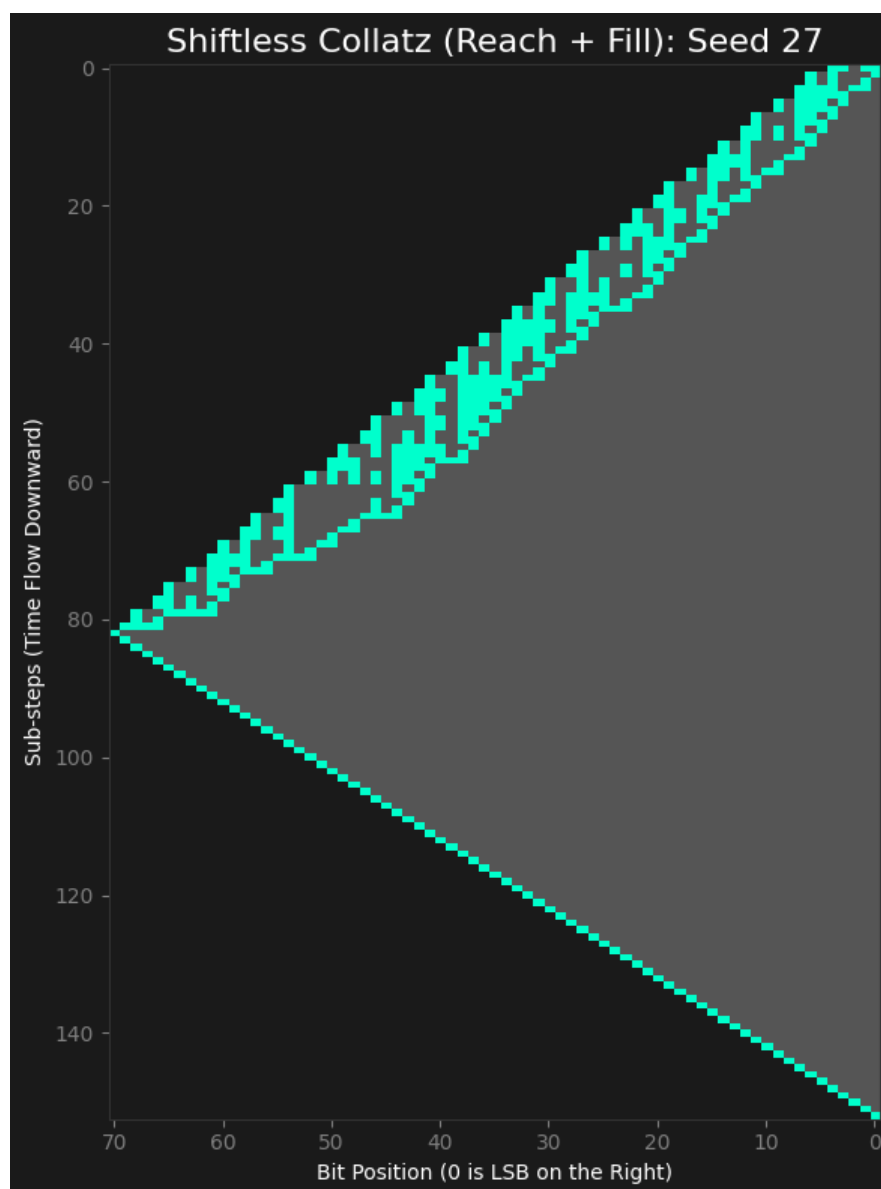


Figure 3. Internal decomposition of a Shiftless Collatz step into Reach ($3n$) and Fill ($3n+LSB$) ($n = 27$).

5. Synchronized Comparison of Three Dynamics

The following three dynamics were compared under the same number of steps:

- $3n$ (pure geometric expansion)
- $3n+1$ (odd-step behavior of the standard Collatz model)
- $3n+LSB$ (Shiftless Collatz model)

From this comparison, the following observations were made (**Figure 4**):

- $3n$ exhibits a clean Sierpiński-gasket structure
- $3n+1$ shows slight distortion but remains globally similar to $3n$
- $3n+LSB$ collapses significantly due to the heavy LSB injection, producing interference fringes

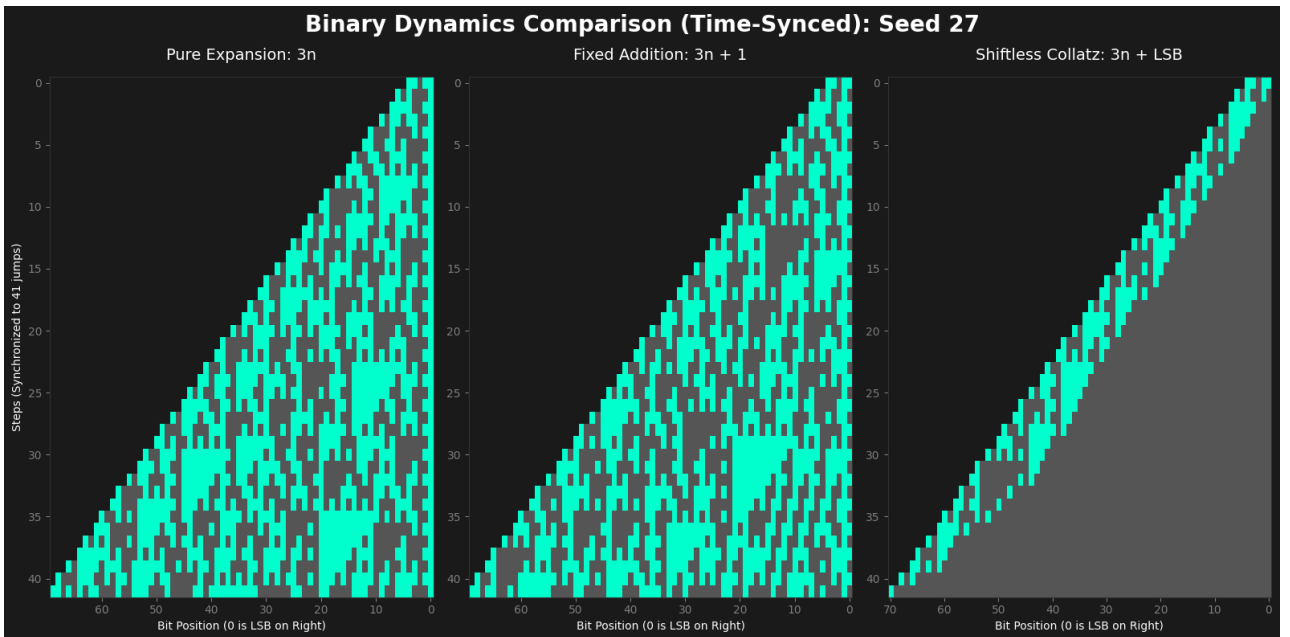


Figure 4. Synchronized comparison of the three dynamics: $3n$, $3n+1$, and $3n+LSB$ ($n = 27$).

6. Linear Decomposition and Interference Geometry

The Shiftless Collatz trajectory can be decomposed into two linear components:

$$\bullet \quad n_k = 3^k \cdot n_0 + \sum 3^{(k-1-i)} \cdot L_i = A_k + B_k$$

Component A (initial seed contribution):

$$\bullet \quad A_k = 3^k \cdot n_0$$

Component B (cumulative LSB contribution):

$$\bullet \quad B_k = \sum 3^{(k-1-i)} \cdot L_i$$

Interference (carry-trigger points):

When A_k and B_k both have a 1 in the same bit position, addition produces a carry, causing the actual bit in n_k to vanish. This explains the fractal structure of the Shiftless Collatz model as a moiré-like interference pattern (**Figure 5**).

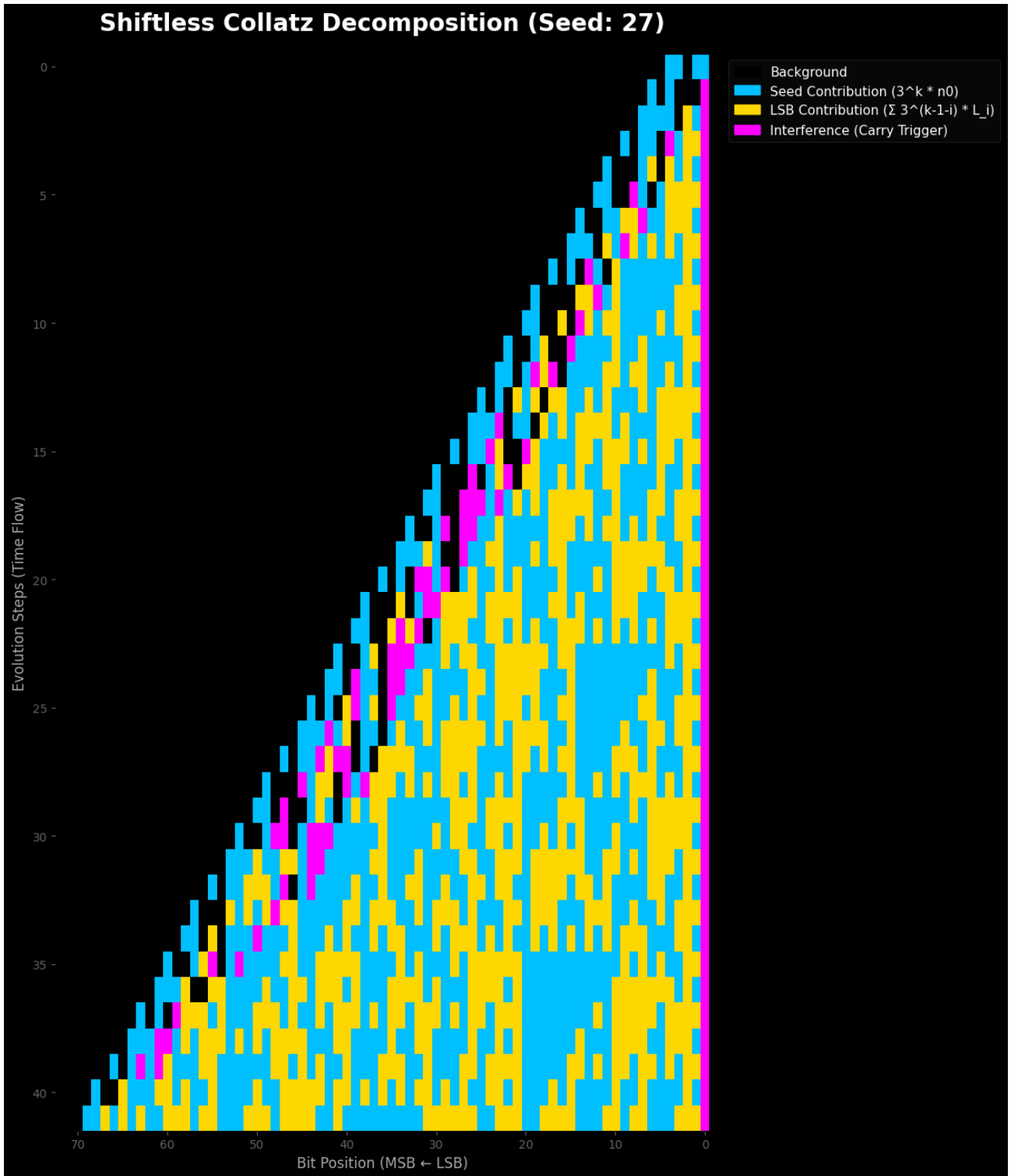


Figure 5. Bitwise visualization of the linear decomposition of n_k into A_k and B_k ($n = 27$).

7. Conclusion

The Shiftless Collatz model is mathematically equivalent to the standard Collatz model, yet it preserves all bit-level information, enabling direct visualization of the internal structure of the trajectory.

This study demonstrates:

- Perfect synchronization of odd parts
- Emergence of fractal structures
- Linear decomposition and interference geometry

The Shiftless Collatz model does not resolve the Collatz conjecture itself, but it provides a powerful visual and mathematical framework for understanding the internal dynamics of the $3n+1$ process.

8. License

Research document: CC BY 4.0

Python source code: MIT License

Copyright (c) 2026 Hiroshi Harada

9. References

- [1] Lagarias, J. C. (1985). *The $3x+1$ problem and its generalizations*.
- [2] Terras, R. (1976). *A stopping time problem on the positive integers*.
- [3] Lagarias, J. C. (2010). *The Ultimate Challenge: The $3x+1$ Problem*.